

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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Github Repository	Click Here

1 Aim and Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition (HAR) dataset, namely:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

The clusters formed by each algorithm are visualized and compared, and their performance is analyzed using internal and external evaluation metrics.

Dataset Description

- **Dataset:** Human Activity Recognition Using Smartphones Dataset.
- **Subjects:** 30 volunteers (aged 19–48).
- **Activities (6 classes):** WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING.
- **Sensors:** 3-axis accelerometer and gyroscope sampled at 50 Hz.
- **Feature Extraction:** Signals were pre-processed into fixed-width sliding windows of 2.56 s (128 readings/window) with 561 time- and frequency-domain features extracted per window.
- **Samples:** The provided train and test splits were concatenated to give 10,299 samples $\times$ 561 features (plus the activity label).

2 Preprocessing Steps

- The pre-extracted train and test feature matrices (`X_train.txt`, `X_test.txt`) and their corresponding labels (`y_train.txt`, `y_test.txt`) were loaded and concatenated into a single dataset of 10,299 samples.
- Feature names were assigned from `features.txt`, with duplicate names disambiguated by appending a numeric suffix.

- Numeric activity labels were mapped to their string names (WALKING, SITTING, etc.) using `activity_labels.txt`.
- **Missing values:** None were found; rows with a missing target were checked for and none existed.
- **Constant / duplicate columns:** Columns with a single unique value and duplicate rows were checked for and removed (none were present in this dataset, so the shape remained $10,299 \times 562$ after preprocessing).
- **Encoding:** The categorical `Activity` target was label-encoded using `LabelEncoder` (retained separately as `ytrue` for external evaluation; the encoded target column itself is excluded from the feature matrix used for clustering).
- **Feature scaling:** All numeric features were standardized using `StandardScaler` (zero mean, unit variance), which is essential since K-Means, DBSCAN and HAC are all distance-based algorithms.

Exploratory Data Analysis

Since clustering is unsupervised, EDA was limited to visualizing the distribution of the ground-truth activity labels (used later only for external validation, not for training).

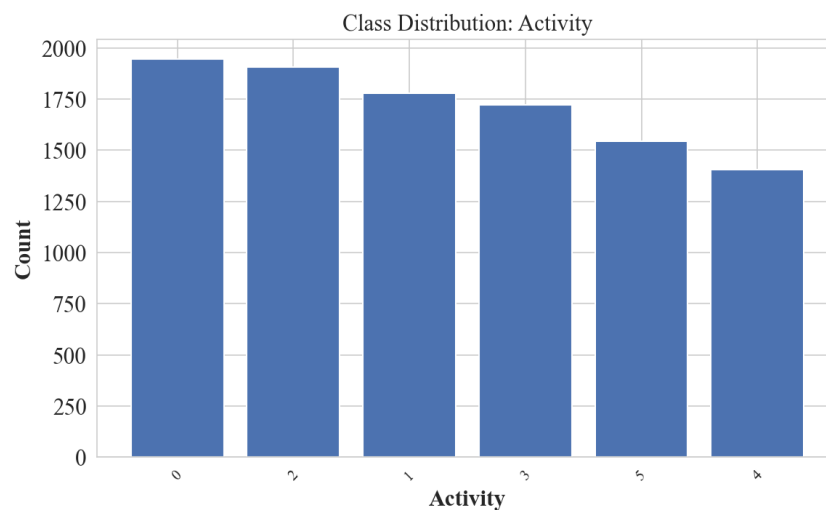


Figure 1: Class distribution of the six activity labels in the HAR dataset.

Inference: The six activity classes are reasonably balanced, with each activity contributing roughly 1,400–1,950 samples. This near-uniform distribution means clustering algorithms are not biased toward a dominant class purely due to sample imbalance, so any imbalance seen in the resulting clusters can be attributed to the algorithm’s grouping behaviour rather than skewed input data.

3 K-Means Clustering with Elbow Method

K-Means was run for $k = 2$ to $k = 8$, computing the Within-Cluster Sum of Squares (WCSS/inertia) and the Silhouette Score (on a sample of 2,000 points for efficiency) at each k .

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3,272,857	0.3966
3	2,921,078	0.3137
4	2,781,602	0.1537
5	2,654,790	0.1340
6	2,577,180	0.1063
7	2,519,774	0.0767
8	2,465,379	0.0729

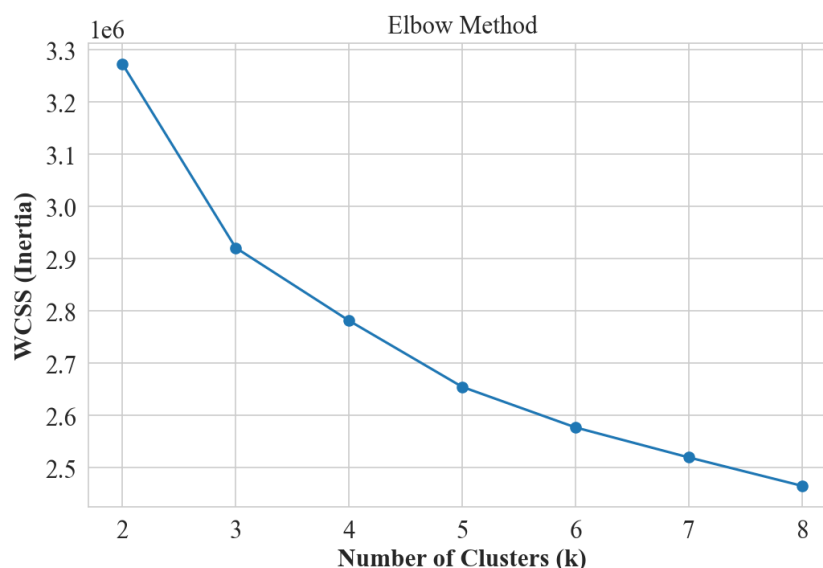


Figure 2: Elbow curve: k vs. WCSS.

Inference: WCSS decreases monotonically with increasing k , as expected. The rate of decrease slows down noticeably after $k = 6$, producing a visible “elbow” in the curve. Beyond this point, adding more clusters yields diminishing reductions in within-cluster variance, indicating that 6 clusters capture most of the useful structure in the data.

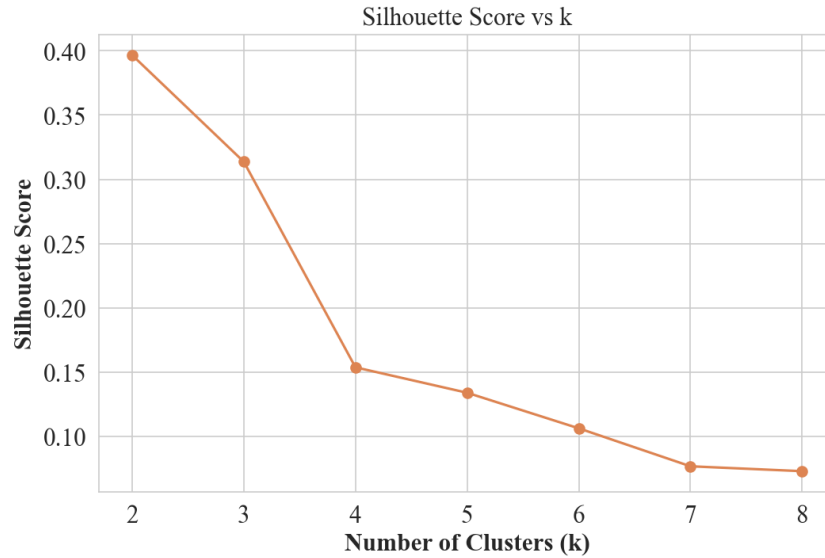


Figure 3: Silhouette Score vs. number of clusters k .

Inference: The silhouette score is highest at $k = 2$ and decreases steadily as k increases, indicating that the data is most separable into two broad clusters (which likely correspond to a “dynamic activity” group vs. a “static activity” group). However, this coarse grouping does not reflect the six real activities present in the ground truth.

Choice of best k : Although the silhouette score alone favours $k = 2$, this experiment selects $k = 6$, since (a) the elbow curve also flattens around $k = 6$, and (b) six is the true number of activity classes in the dataset, which allows a meaningful one-to-one comparison between the discovered clusters and the ground-truth activities during external evaluation. This is a common trade-off in real-world clustering: the purely geometric objective (silhouette/WCSS) can suggest a coarser grouping than the number of semantically meaningful classes in the data.

4 Clustering Implementation

K-Means (Model A)

K-Means was re-fit with the chosen $k = 6$ (`random_state=42`, `n_init=10`) on the standardized feature matrix using Euclidean distance.

DBSCAN (Model B)

To select DBSCAN’s neighbourhood radius ϵ , a k-distance plot was generated using the distance to each point’s 5th nearest neighbour, sorted in ascending order.

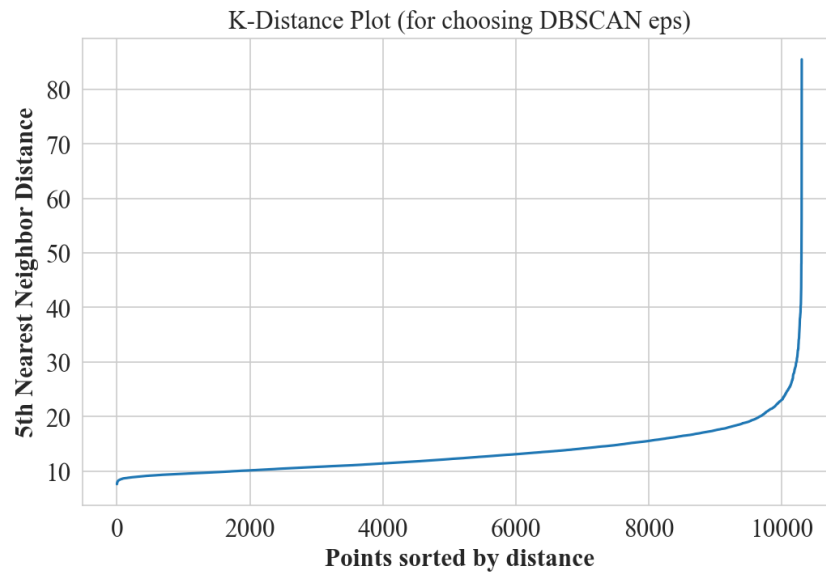


Figure 4: K-Distance plot used to select DBSCAN's ϵ parameter.

Inference: The plot shows a gradual rise followed by a sharp knee. The value of the distance at this knee is a good candidate for ϵ , since points beyond it are considerably farther from their neighbours and are more likely to be noise. Based on this plot, DBSCAN was run with $\epsilon = 15$ and **minPts = 10**.

With these parameters, DBSCAN discovered **5 clusters** and labelled **2,398 out of 10,299 points (about 23.3%)** as noise.

Hierarchical Agglomerative Clustering (Model C)

Agglomerative clustering was performed with **Ward's linkage** and **6 clusters**, matching the number chosen for K-Means so that the two algorithms can be compared fairly.

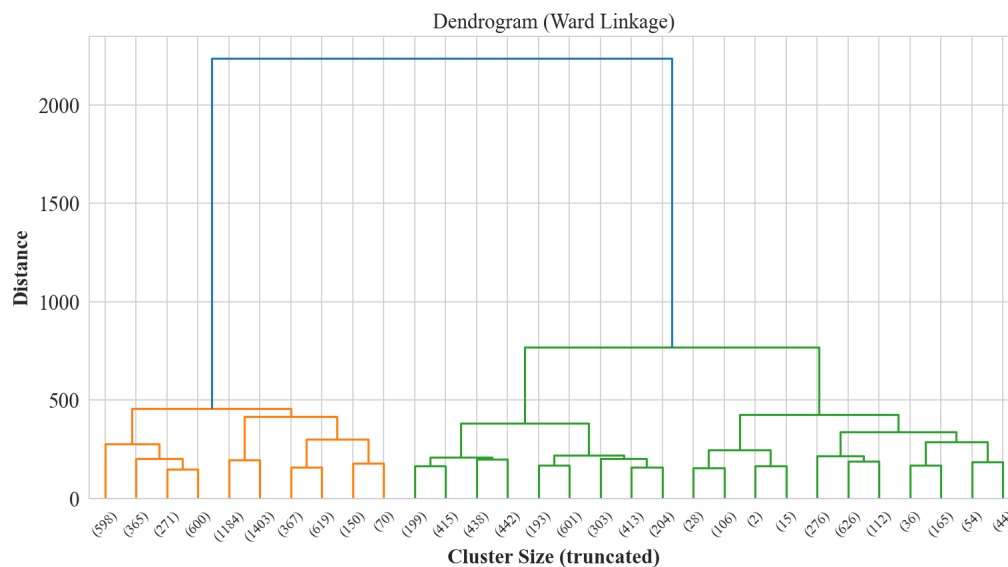


Figure 5: Dendrogram of Hierarchical Agglomerative Clustering using Ward's linkage (truncated to the last 30 merges).

Inference: The dendrogram shows several tight, low-height merges near the leaves (samples of the same activity being very similar to one another) followed by a few large jumps in merge distance higher up the tree. These large jumps indicate natural boundaries between dissimilar groups, and cutting the tree at a height corresponding to 6 clusters produces reasonably well-separated branches.

5 Cluster Visualization

The 561-dimensional feature space was projected to 2D using PCA (and additionally t-SNE, computed on the first 50 principal components) to visually inspect the clusters formed by each algorithm against the ground-truth activity labels.

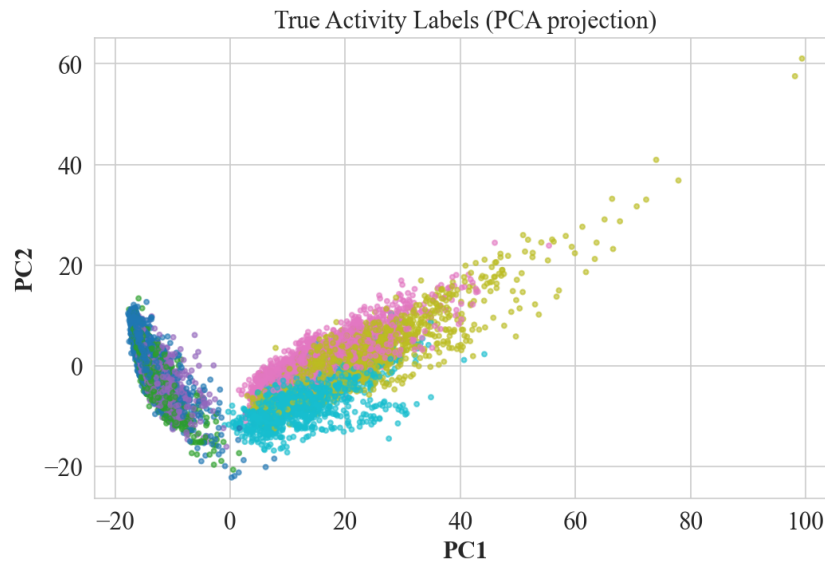


Figure 6: Ground-truth activity labels, PCA projection.

Inference: The PCA projection shows two broad super-clusters: one tight, well-separated group corresponding to static activities (SITTING, STANDING, LAYING) and one more diffuse, overlapping group corresponding to dynamic activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS). This confirms why $k = 2$ maximizes the silhouette score.

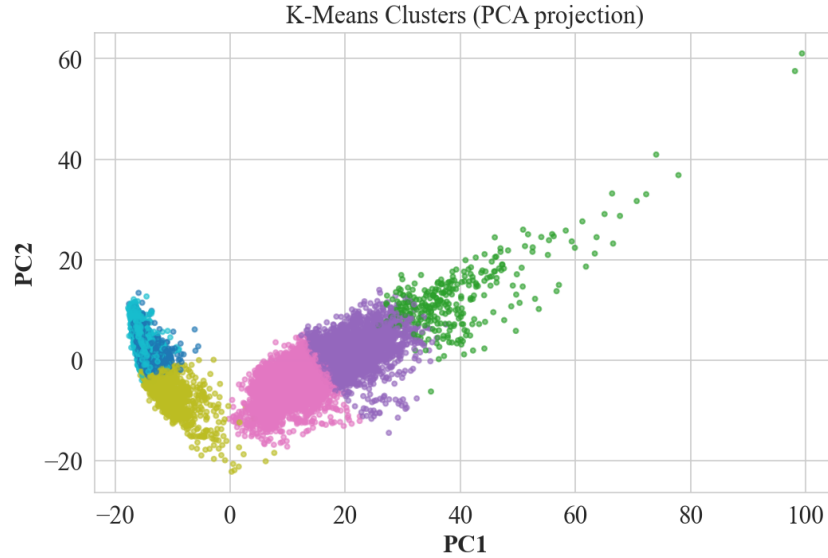


Figure 7: K-Means clusters ($k = 6$), PCA projection.

Inference: K-Means successfully separates the static activities into distinct, compact clusters but struggles to cleanly separate the three dynamic activities from one another, since their feature distributions overlap substantially in the reduced 2D space.

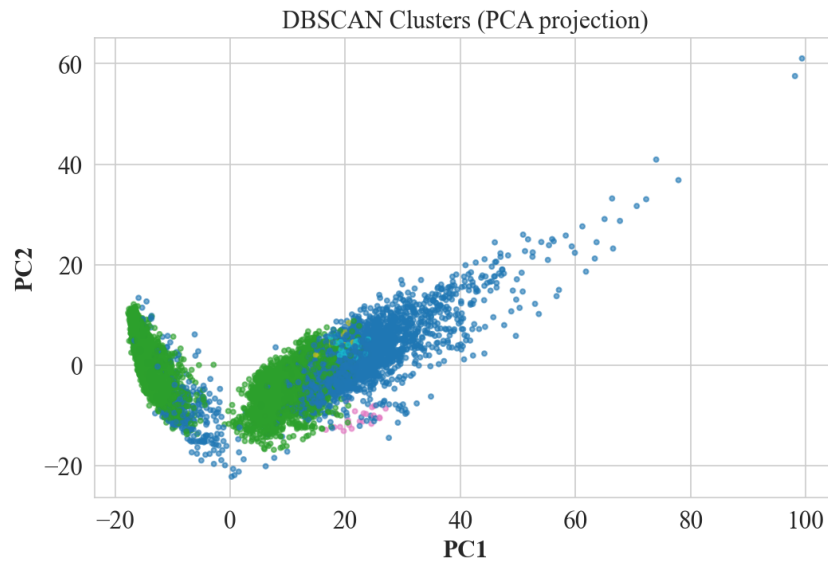


Figure 8: DBSCAN clusters, PCA projection (noise points shown separately).

Inference: DBSCAN correctly identifies the dense static-activity regions as separate clusters but assigns a large fraction of the more diffuse dynamic-activity points to noise, since their local density falls below the threshold implied by $\epsilon = 15$ and $\text{minPts} = 10$. This explains DBSCAN's relatively low external validation scores despite a high silhouette score.

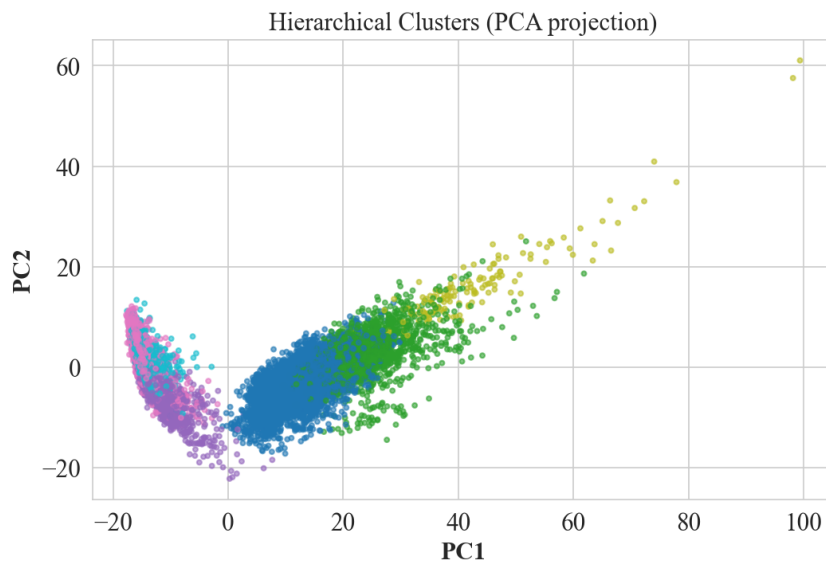


Figure 9: Hierarchical (Ward linkage) clusters, PCA projection.

Inference: The hierarchical clustering result closely resembles the K-Means result for the static activities and offers a marginally better split of the dynamic activities, consistent with its slightly higher ARI and NMI scores reported later.

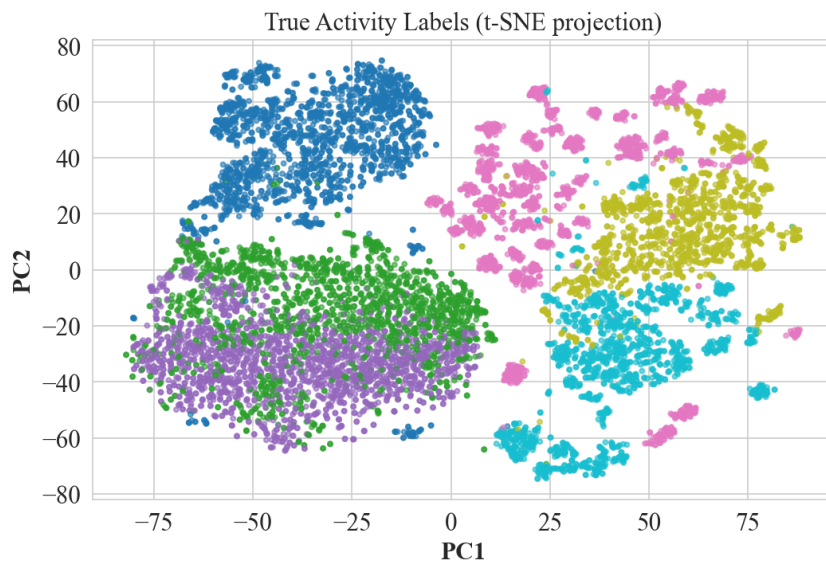


Figure 10: Ground-truth activity labels, t-SNE projection.

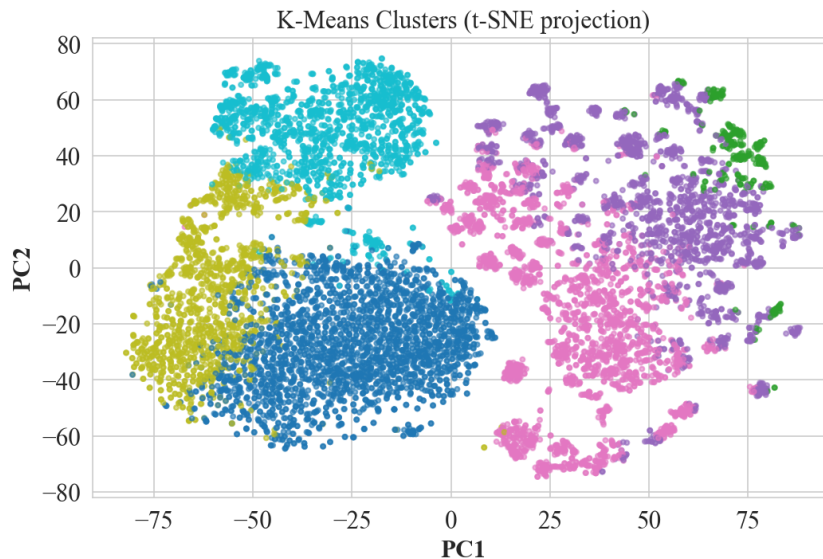


Figure 11: K-Means clusters ($k = 6$), t-SNE projection.

Inference: The t-SNE projection expands the dynamic-activity region into three visually distinguishable sub-clusters, which are not as apparent under PCA. Comparing this against the K-Means t-SNE plot shows that K-Means recovers most of these sub-groups reasonably well, though some boundary points are still misassigned between WALKING and WALKING_DOWNSTAIRS.

6 Evaluation Metrics and Results

Internal Evaluation Metrics

Table 2: Internal Clustering Metrics

Model	Silhouette	Davies–Bouldin	Calinski–Harabasz
K-Means	0.1063	2.3836	2556.54
DBSCAN	0.2280	1.1543	50.65
Hierarchical	0.1110	2.4820	2349.67

External Evaluation Metrics (w.r.t. Ground Truth)

Table 3: External Clustering Metrics

Model	ARI	NMI
K-Means	0.4196	0.5593
DBSCAN	0.0800	0.1538
Hierarchical	0.4599	0.6015

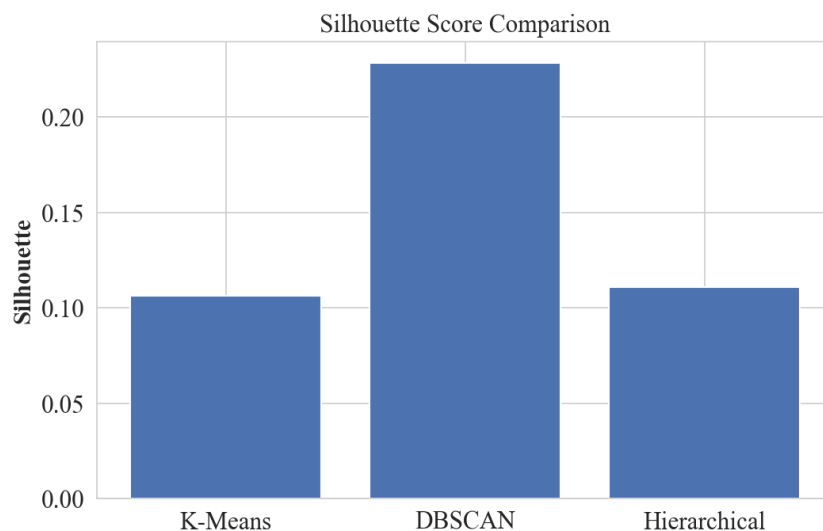


Figure 12: Silhouette score comparison across the three clustering models.

Inference: DBSCAN achieves the highest silhouette score, but this is misleading: silhouette scores are computed only over non-noise points, so DBSCAN is effectively being scored on the subset of “easy”, well-separated points it chose to cluster, while ignoring the 23.3% of points it labelled as noise.

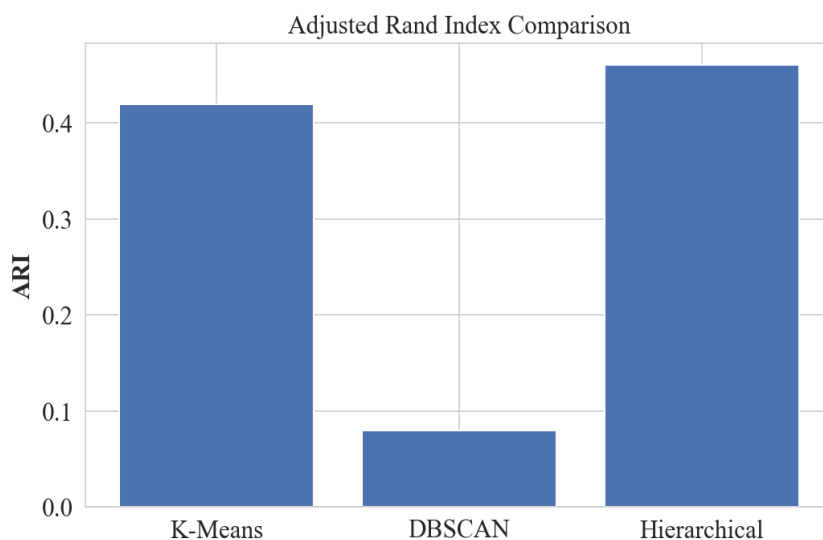


Figure 13: Adjusted Rand Index (ARI) comparison across the three clustering models.

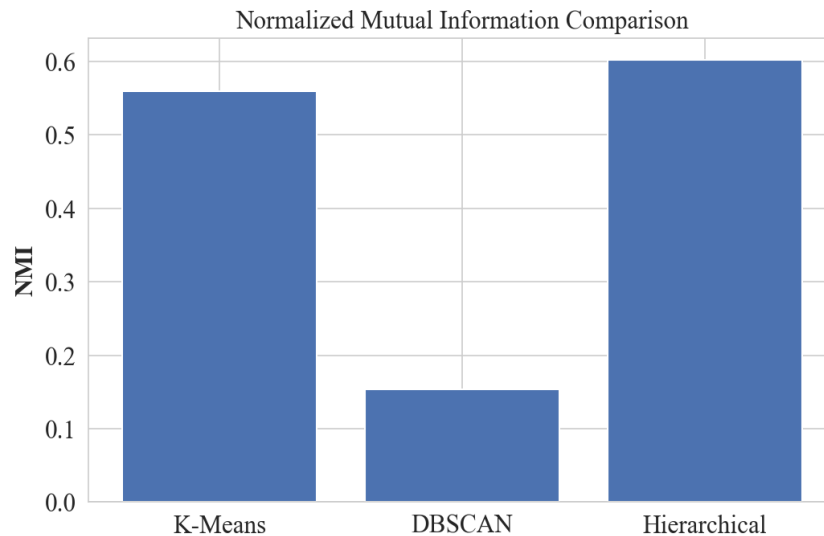


Figure 14: Normalized Mutual Information (NMI) comparison across the three clustering models.

Inference: On both external metrics (ARI and NMI), which measure agreement with the true activity labels, Hierarchical Clustering performs best, followed closely by K-Means, while DBSCAN performs considerably worse. This confirms that DBSCAN’s high internal (silhouette) score does not translate into better agreement with the actual activity classes, since it discards a large fraction of the harder-to-cluster dynamic-activity points as noise instead of assigning them to a class.

7 Conclusion

K-Means and Hierarchical Agglomerative Clustering (with Ward’s linkage) produced clusters that agree reasonably well with the true HAR activity labels ($\text{ARI} \approx 0.42\text{--}0.46$, $\text{NMI} \approx 0.56\text{--}0.60$), with Hierarchical Clustering marginally outperforming K-Means on both external metrics. DBSCAN, while achieving the best internal (silhouette) score, performed worst on external validation because it excludes a large proportion of harder-to-separate dynamic-activity samples as noise rather than clustering them. Overall, the static activities (SITTING, STANDING, LAYING) are consistently well-separated by all three algorithms, while the dynamic activities (WALKING and its variants) remain difficult to distinguish using unsupervised clustering alone.